

A PROJECT WITH



n8n

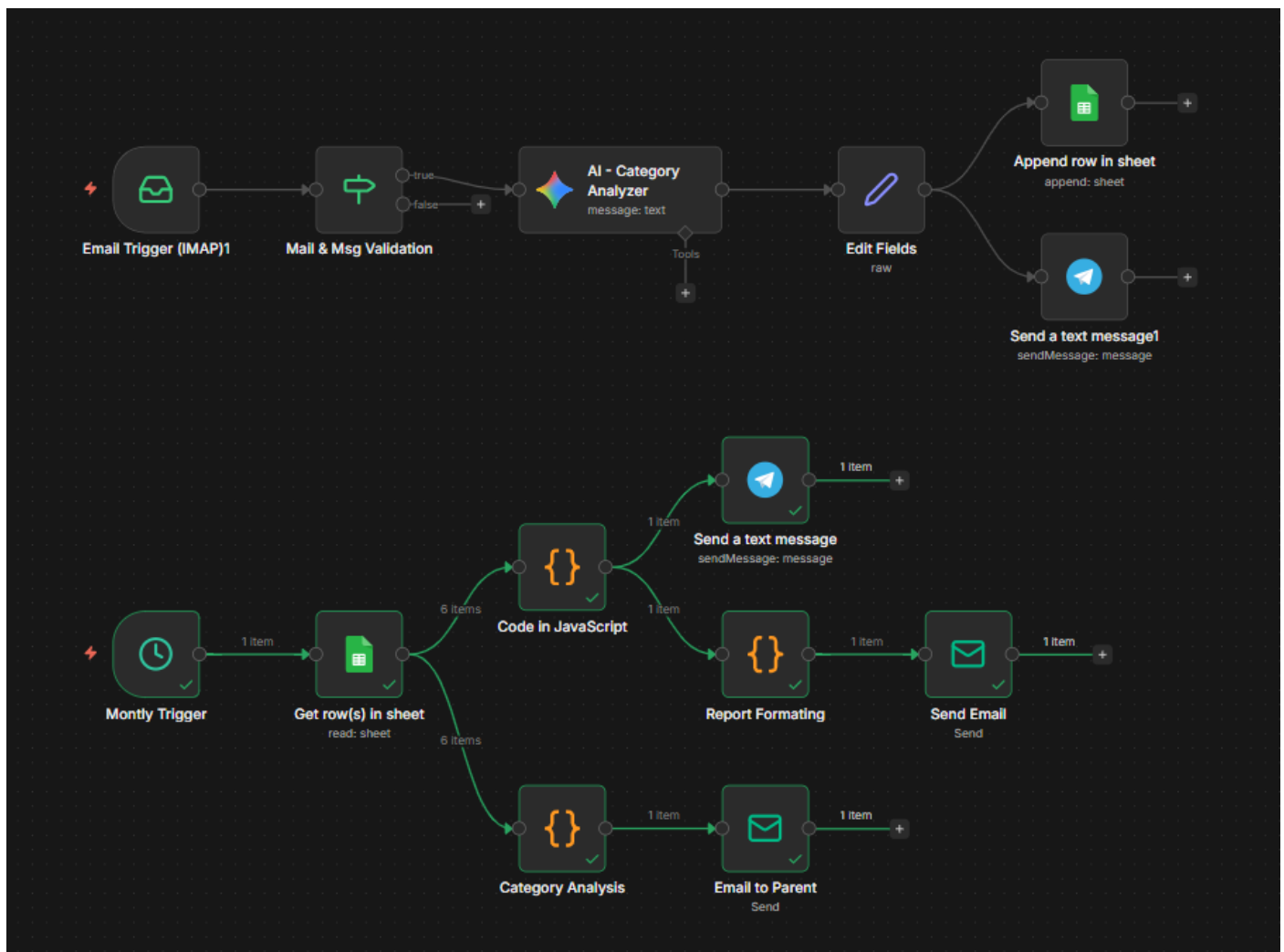


EXPENSE TRACKING ANALYSIS (USING N8N)

Project Objective:

- Expense logging via Email
- AI-based categorization
- Structured data storage
- Real-time Telegram notification
- Monthly financial analytics report
- HTML + graphical reporting via Email

All without manual intervention.



Tech Stack Used:

- n8n (Docker containerized)
- Gmail App Password for authentication
- Google Console API & Services
 - o OAuth2 token authentication (Enabled Google Sheets & Drive)
- AI Layer
 - o Gemini AI API Token Based integration
- Messaging
 - o Telegram Bot + Chat ID Integration

System Architecture:

The project contains two independent automation flows:

Flow 1:

- User sends an email to Gmail.
- Subject must contain the word **“Expense”**
- Send message to Gemini AI & categorizes the expense
- Format structured data
- Append row into Google Sheet
- Send Telegram confirmation

Flow 2: (Monthly Cron Job)

- Monthly Trigger
- Fetch Sheet Data & Filter current month
- JavaScript analysis:
 - o Total spends
 - o Category-wise totals
- Format structured HTML report
- Send
 - o Telegram summary
 - o Detailed Email HTML Report

Design Considerations:

- Hardcoded for single-user use
- Email and Telegram IDs fixed
- No multi-user authentication
- No database (used Google Sheets as storage)
- No public deployment (local Docker)

Samples Images:

